SAMPLE

History of Astronomy: The Big Bang Theory

From Idea to Consensus

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Free sample: Introduction & Chapter 1

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Introduction

Every story about the universe has to answer one question before all others: did it have a beginning? For most of recorded history, that question belonged to philosophers and theologians rather than scientists, because no telescope could peer far enough back to settle it. That changed during the twentieth century. Astronomers measured the motions of distant galaxies, physicists rebuilt the laws of gravity, and a strange new picture emerged: the universe is not eternal and unchanging. It has a history. It was once far hotter, far denser, and far smaller than it is today, and it has been expanding and cooling ever since.

That picture is the Big Bang theory, and this book tells the story of how it won. It is a history of astronomy, not a physics textbook, so the spotlight falls on people: a Belgian priest who insisted that the universe had a beginning when almost nobody agreed with him, an American lawyer turned astronomer who measured the expanding universe from a mountaintop in California, a stubborn British theorist who gave the theory its mocking name and then spent decades fighting it, and two radio engineers in New Jersey who accidentally discovered the faint afterglow of the young cosmos while debugging a noisy antenna.

The Big Bang theory was not accepted overnight. It arrived in pieces, argued over half a century. The first piece was expansion, established in the 1920s. The second was the hot dense beginning, proposed in 1931 and hardened into working physics in the 1940s. The third was the fossil radiation left over from the early universe, predicted in 1948 and found, to everyone's astonishment, in 1965. Each piece faced a serious rival, most famously the steady state model, which held that the universe had no beginning at all. The rivalry was resolved not by rhetoric but by evidence: radio galaxies, quasars, and above all the cosmic microwave background, a whisper of static coming from every direction of the sky that no eternal, unchanging universe could explain.

Understanding this history matters for more than trivia. The Big Bang theory is the first scientific account of cosmic origins that rests on measurement rather than myth, and the way it was established -- prediction, rival, test, verdict -- is one of the clearest demonstrations of the scientific method at work. It also changed the questions scientists ask. Once the universe had a beginning, it became something with an age, a childhood, and a possible end, and cosmology grew from a speculative backwater into one of the most productive fields in all of science.

This book walks through that transformation one step at a time. Early chapters cover the observations that forced astronomers to abandon an unchanging cosmos, and explain in plain language what the theory actually claims -- because, despite its name, the Big Bang was not an explosion of stuff into empty space. Middle chapters cover the great debates and the detective work: how theorists calculated the universe's early chemistry, how skeptics proposed alternatives, and how new instruments delivered verdicts. Later chapters bring the story to the present, from the satellites that mapped the afterglow of creation to the deep-space telescopes now photographing the universe's first galaxies, and the puzzles that keep cosmologists awake at night, such as the stubborn disagreement between two ways of measuring how fast the universe expands.

No prior physics is needed. Where an idea needs a number, the number is explained; where it needs an analogy, the analogy is honest. The reward for the journey is a rare one: by the final chapter you will know where everything you can see came from, how we know it, and how recently our species figured it out. The universe is about 13.8 billion years old, yet the certainty of that very recent statement is barely sixty years old. This is the story of how that certainty was earned, and of the people who earned it against their own expectations.

Chapter 1: The Discovery of The Big Bang Theory

The discovery that the universe is expanding began, oddly enough, with a farm in Kansas. Vesto Slipher, an astronomer at Lowell Observatory in Flagstaff, Arizona, spent the 1910s photographing the spectra of faint, fuzzy patches of light called spiral nebulae. Using the observatory's 24-inch refractor, he painstakingly measured how their light was shifted along the spectrum. Slipher found something startling: nearly all of them had light stretched toward the red end, meaning they were racing away from us at tremendous speeds. One nebula, later known as the Andromeda Galaxy, was approaching, but the great majority were receding, some at over a thousand kilometers every second. By 1917 Slipher had logged more than a dozen of these redshifts, and he reported his results quietly at meetings. He had gathered the first evidence that the cosmos is not standing still, but he did not know what the nebulae actually were, and he did not pursue the implications. Sometimes a discovery waits for the right person to ask the right question of it.

The right questions were being asked, meanwhile, in the world of theory. In 1915 Albert Einstein published his

general theory of relativity, which recast gravity as the curvature of space and time. When Einstein applied his new equations to the universe as a whole, he ran into a problem: the equations refused to describe a static cosmos. Matter curves space, and curved space pulls matter together, so an unmoving universe ought to collapse. Since Einstein, like nearly everyone, believed the universe was eternal and changeless, he added a fudge factor to his equations -- the cosmological constant -- to hold the cosmos up. He would later call this the biggest blunder of his career, though the story of that remark is more anecdote than record.

Others were bolder. In 1922 the Russian mathematician Alexander Friedmann solved Einstein's original equations without the fudge factor and showed that they naturally described universes that expand or contract with time. Einstein initially dismissed Friedmann's work, then retracted the dismissal after Friedmann protested, but the paper still went mostly unread in the West. Five years later the Belgian priest and physicist Georges Lemaître reached the same conclusion independently, published it in an obscure Belgian journal, and met with the same silence. In 1929, unaware of Friedmann and Lemaître, the American cosmologist Edwin Hubble published the observation that changed everything.

Hubble was a lawyer turned astronomer with a taste for grand claims, working at Mount Wilson Observatory above Pasadena with the largest telescope on Earth, the 100-inch Hooker reflector. His great contribution was twofold. First, with Milton Humason's patient spectroscopy building on Slipher's program, he assembled redshifts for galaxies far beyond Slipher's reach. Second, he proved that the spiral nebulae were not clouds within our own Milky Way but immense star systems in their own right, island universes millions of light-years away. That distance measurement mattered because it let Hubble compare how fast galaxies were moving with how far away they sat. The pattern he found was simple and shocking: the farther a galaxy was, the faster it was receding. Distance and speed rose together, roughly doubling distance doubled the recession speed.

That correlation is exactly what expansion predicts. It is not that galaxies are fleeing from us through space like shrapnel from a bomb, which would produce no such tidy relation. It is that the space between galaxies is stretching, and light traveling through that stretching space gets stretched with it, reddening it. A galaxy twice as far away has twice as much expanding space between us, so its light is stretched twice as much. Hubble had discovered, though he resisted the interpretation for years, the expanding universe. Astronomers did not immediately leap to a cosmic beginning, but the seed of the Big Bang theory was planted: if you run an expanding universe backward in time, everything gets closer together, hotter, and denser. Keep going, and there is a first moment.

Two details complete the discovery. Hubble's original numerical value for the expansion rate was far too high, about seven times the modern figure, because of calibration errors in his distance measurements; the errors were corrected over the following decades. And the person who finally connected Hubble's law to a cosmic beginning was Lemaître. In 1931, rereading his own 1927 solution and Hubble's observations together, Lemaître proposed that the universe began from a single initial state he called the primeval atom -- a hot, dense starting point from which all of space, time, and matter had expanded. It was the first published version of what would be called the Big Bang. Few paid attention. One who did was Einstein, who by then had abandoned his cosmological constant and told Lemaître, reportedly while standing on a California terrace, that the worst thing about the idea was that it was so simple he had not thought of it himself.